

New Constructions of Multiplicative Secret Sharing Schemes

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Abstract

This paper investigates the multiplicative properties of linear codes in secret sharing schemes. To address the limitation that certain access structures cannot be realized by ideal linear codes, we introduce the notion of shortest linear codes as an ideal benchmark for code length. Since explicitly determining such shortest codes is generally computationally difficult, we propose an explicit construction that, for any given access structure, produces a length-efficient linear code whose induced access structure. On this basis, we further define multiplicative ideal linear codes and multiplicative length-efficient linear codes, and derive necessary and sufficient conditions for the existence of multiplicativity. The effectiveness of the proposed approach is demonstrated by concrete examples. Compared with the construction of Cramer *et al.*, the multiplicative linear codes obtained in this work have smaller length.

Index Terms

Secret sharing schemes, Linear codes, Shortest linear codes, Length-efficient constructions, Multiplicativity.

I. INTRODUCTION

SECRET sharing schemes are fundamental primitives in cryptography and distributed computation. The core idea is to distribute a secret s among a set of participants $P = \{p_1, p_2, \dots, p_n\}$ in such a way that only certain authorized subsets of participants can reconstruct the secret, while all unauthorized subsets obtain no information about it. The collection of all authorized subsets is called an access structure Γ . Secret sharing has been extensively studied since the seminal works of Blakley [1] and Shamir [2], and has found numerous applications, in particular as a basic building block for secure multiparty computation (MPC) protocols.

Homomorphic secret sharing schemes were first proposed by Benaloh [3]. Such schemes allow participants to perform computations directly on their shares without revealing them, in such a way that the result is equivalent to first applying the corresponding operation to the secret and then sharing it. According to whether the sharing function is additive or multiplicative, one distinguishes between additively homomorphic and multiplicatively homomorphic secret sharing schemes. Compared with the former, the latter is substantially more difficult to construct.

Monotone span programs (MSP) provide a powerful linear-algebraic framework for representing and analyzing secret sharing schemes. It is well known that many constructions for general access structures can be phrased in terms of MSP, and that MSP-based schemes capture a broad class of linear secret sharing constructions [4] [5]. In the MSP framework, criteria for multiplicativity have been established, and explicit constructions of multiplicative secret sharing schemes are available for a wide range of access structures, e.g., Cramer *et al.* [6]. In particular, given a monotone span program, one can decide whether the corresponding scheme is multiplicative and, under suitable conditions, derive multiplicative constructions.

However, for secret sharing schemes based directly on linear codes—including classical schemes such as Shamir's secret sharing scheme [2], which is closely related to Reed–Solomon codes [7]—no general criterion is available to check multiplicativity for an arbitrary access structure from a given linear code, and no explicit procedure is known to construct a multiplicative linear code for a given access structure. Although the relationship between linear codes and secret sharing schemes has been widely explored [8] [9] [10] [11], many existing works build secret sharing schemes from specific families of linear codes and therefore address only the corresponding families of access structures (e.g., [9] [10]), and do not address, in a general and constructive way, the problem of deciding and achieving multiplicativity in the linear-code setting. This gap motivates the present work.

In the linear-code-based approach, one typically starts from an $[n + 1, k]$ linear code C over F_q with generator matrix $G = (g_0^T, g_1^T, \dots, g_n^T)$, where the first column g_0^T corresponds to the secret and the remaining columns correspond to the participants' shares. The induced secret sharing scheme realizes an access structure $\Gamma(C)$ determined by the supports of certain codewords in the dual code C^* . If for a given target access structure Γ there exists a linear code C such that $\Gamma(C) = \Gamma$ and all shares have the same size as the secret, C is called an ideal linear code for Γ . In the broader secret sharing literature,

ideal secret sharing schemes have been extensively studied [12] [13]. However, Benaloh and Leichter showed that there exist access structures for which no secret-sharing scheme realizing the structure can be ideal [14]; consequently, no ideal linear code exists, and one cannot restrict attention to ideal schemes.

To address this limitation, we introduce the definition of shortest linear codes, which minimize the length among all linear codes realizing a given access structure Γ . Since explicitly determining such shortest codes is generally computationally difficult, we propose an explicit construction over F_q that starts from a parity-check matrix derived from the minimal authorized sets of Γ , and then adds carefully selected candidate columns to obtain a new linear code whose induced access structure coincides with the target Γ while having comparatively small length. In this way, for any access structure, our construction yields a length-efficient linear code realizing Γ , even when no ideal linear code exists or when constructing one is difficult. Building on this framework, we further define multiplicative ideal linear codes and multiplicative length-efficient linear codes, derive necessary and sufficient conditions for multiplicativity in terms of explicit linear systems, and demonstrate through concrete examples that the resulting multiplicative linear codes have smaller length than those obtained from the classical monotone-span-program-based construction of Cramer *et al.* [6].

The remainder of this paper is organized as follows. Section II reviews basic concepts of linear codes and their connection to secret sharing schemes. Section III discusses multiplicative ideal linear codes and provides a characterization of multiplicativity. Section IV introduces shortest linear codes as an ideal length benchmark and proposes an explicit construction of length-efficient linear codes realizing arbitrary access structures. Section V studies the multiplicativity of these length-efficient linear codes and presents explicit examples comparing our constructions with those of Cramer *et al.*. Section VI concludes the paper and outlines directions for future work.

II. PRELIMINARIES

A. Linear Codes

To investigate the connection between secret sharing schemes and linear codes, it is first necessary to understand some fundamental concepts and related results concerning linear codes [15]. Let $\mathbb{F}_q^{n+1}$ denote the $(n+1)$ -dimensional vector space over the finite field $\mathbb{F}_q$. A linear code C of parameters $[n+1, k]$ is a k -dimensional subspace of the vector space F_q^{n+1} . Each element $c = (c_0, c_1, \dots, c_n) \in C$ is called a codeword, and by selecting k linearly independent codewords from C , a matrix G of size $k \times (n+1)$ is formed, with the selected codewords as its row vectors. The matrix G is called the generator matrix of the linear code C , because the row vectors of G span exactly the subspace C , i.e., $c = \{uG | u \in F_q^k\}$.

For vectors $v, w \in \mathbb{F}_q^{n+1}$, the notation $\langle v, w \rangle$ denotes the inner product of the vectors. The dual code of the linear code C is defined as $C^* = \{c' \in \mathbb{F}_q^{n+1} | \forall c \in C, \langle c, c' \rangle = 0\}$. It follows that if the generator matrix of the linear code C is G , then $C^* = \{v \in \mathbb{F}_q^{n+1} | vG^T = 0\}$, and C^* is a linear code of parameters $[n+1, n+1-k]$ over $\mathbb{F}_q$. Let H be an $(n+1-k) \times (n+1)$ matrix, and if $GH^T = 0$, then H is called the parity-check matrix of the linear code C . Based on the properties of linear codes, it is known that the parity-check matrix of C is the generator matrix of the dual code C^* .

B. Secret Sharing From Linear Codes

The following describes a method for constructing a secret sharing scheme based on linear codes [8]: Let C be a linear code of parameters $[n+1, k]$, and G be the generator matrix of C , where $G = (g_0^T, g_1^T, \dots, g_n^T)$, with each $g_i^T \in \mathbb{F}_q^k$ being a column vector of G , and $0 \leq i \leq n$. Let $P = \{p_1, p_2, \dots, p_n\}$ be the set of n participants, and $s \in \mathbb{F}_q$ be the secret. First, the secret management center randomly selects $u \in \mathbb{F}_q^k$, such that $ug_0^T = s$, and computes $uG = (s_0, s_1, \dots, s_n)$. Clearly, $uG = (s_0, s_1, \dots, s_n)$ is a codeword of the linear code C , and $s_0 = s$. Then, the secret management center secretly transmits s_i to participant P_i as their share, for $0 \leq i \leq n$. This completes the distribution of the secret, thereby constructing a secret sharing scheme. For simplicity, hereafter, when referring to the “secret sharing scheme obtained from the linear code C ,” it refers to the scheme constructed using the method described here.

Let $A = \{P_{i_1}, \dots, P_{i_m}\} \subseteq P$ where $1 \leq i_1 < \dots < i_m \leq n$. The members of A can recover the secret s from their shares $s_{i_1}, \dots, s_{i_m}$ if and only if g_0 can be written as a linear combination of $g_{i_1}, \dots, g_{i_m}$. Let the secret sharing scheme obtained above implement the access structure Γ , then $\Gamma = \{A \subseteq P | g_0 \in \text{span}\{g_i | P_i \in A\}\}$.

The following lemma gives a dual-code characterization of the access structure implemented by the above linear-code-based secret sharing scheme.

Lemma 1: [8] Let G be the generator matrix of the linear code C over $\mathbb{F}_q$ with parameters $[n+1, k]$, and let the secret sharing scheme derived from the linear code C implement the access structure Γ . For a subset of participants $A = \{P_{i_1}, \dots, P_{i_m}\} \subseteq P$, where $1 \leq i_1 < \dots < i_m \leq n$, we have $A \in \Gamma$ if and only if there exists a codeword in the dual code C^* of C of the form

$$(1, 0, \dots, 0, c_{i_1}, 0, \dots, c_{i_m}, 0, \dots, 0),$$

where at least one of $c_{i_1}, \dots, c_{i_m}$ is nonzero.

C. Ideal Linear Code

Let $\mathbf{c} = (c_0, c_1, \dots, c_n)$ be a codeword in the linear code C , and define the support of $\mathbf{c}$ as $\text{supp}(\mathbf{c}) = \{i \in [0, n] \mid c_i \neq 0\}$. Let A be a subset of $P = \{P_1, \dots, P_n\}$. If $\mathbf{c} = (c_0, c_1, \dots, c_n) \in C^*$ with $c_0 = 1$ and $\text{supp}(\mathbf{c}) \subseteq \text{supp}(A) \cup \{0\}$, then A is called an authorized subset of the linear code C . If A is an authorized subset, then any participant set that contains A is also an authorized subset. If every proper subset of an authorized subset A is not an authorized subset, then A is called a minimal authorized subset. The set $\Gamma(C)$ denotes the collection of all minimal authorized subsets in the linear code C . Clearly, a set A is an authorized subset of the linear code C if and only if A contains some element of $\Gamma(C)$.

Suppose $\Gamma = \{S_1, S_2, \dots, S_m\}$ is a collection of subsets of the participant set $P = \{P_1, \dots, P_n\}$, and that no two elements of Γ contain one another. Then Γ defines an access structure, and a subset $A \subseteq P$ is an authorized subset if and only if A contains at least one element of Γ . In other words, Γ is precisely the collection of all minimal authorized subsets.

Definition 1 (Ideal linear code): If there exists a linear code C of parameters $[n+1, k]$ such that $\Gamma(C) = \Gamma$, then the linear code C is called an *ideal linear code* for Γ .

III. MULTIPLICATIVE IDEAL LINEAR CODES

First, we analyze how the participants can securely compute $s + s'$ using the secret-sharing scheme derived from the linear code C . It can be shown that there exist $u, u' \in \mathbb{F}_q^k$ such that

$$\begin{aligned} s + s' &= ug_0^\tau + u'g_0^\tau \\ &= u \left(\sum_{i=1}^n a_i g_i^\tau \right) + u' \left(\sum_{i=1}^n a_i g_i^\tau \right) \\ &= \sum_{i=1}^n a_i s_i + \sum_{i=1}^n a_i s'_i \\ &= \sum_{i=1}^n a_i (s_i + s'_i). \end{aligned}$$

This equation shows that the participants do not need to disclose their original shares s_i and s'_i . Instead, they only need to locally compute the sums $s_i + s'_i$, after which $s + s'$ can be reconstructed by an authorized set through a suitable linear combination. The fact that A is an authorized set guarantees the existence of the coefficients a_i .

However, for the multiplication ss' , the situation becomes more intricate:

$$\begin{aligned} ss' &= (ug_0^\tau)(u'g_0^\tau) \\ &= \left(\sum_{i=1}^n a_i s_i \right) \left(\sum_{j=1}^n a_j s'_j \right) \\ &= \sum_{i=1}^n \sum_{j=1}^n a_i a_j s_i s'_j. \end{aligned}$$

This expansion includes cross terms such as $s_i s'_j$ for $i \neq j$. To ensure security, it is essential to prevent potential information leakage arising from the multiplication of shares held by different participants. Consequently, we require that the multiplication computation rely solely on the local products of each participant's own shares. To address this requirement, we introduce the notion of a multiplicative ideal linear code.

Definition 2 (Multiplicative ideal linear code): For any access structure Γ , suppose there exists a linear code C of parameters $[n+1, k]$ such that $\Gamma(C) = \Gamma$, i.e., the linear code C is ideal. If there exists a vector of recombination coefficients $Z = (z_1, \dots, z_n)$ such that, for all $s, s' \in \mathbb{F}_q$,

$$ss' = \sum_{i=1}^n z_i s_i s'_i,$$

then the ideal linear code C is called *multiplicative*.

The following theorem provides an explicit algebraic criterion for multiplicativity in terms of the solvability of a linear system over $\mathbb{F}_q$.

Theorem 1: Let C be an ideal linear code of parameters $[n+1, k]$ over F_q with generator matrix G . Then C is a multiplicative ideal linear code if and only if the linear system of equations in the unknowns $z_1, \dots, z_n$

$$(G \setminus g_0^\tau) \begin{pmatrix} z_1 & 0 & \cdots & 0 \\ 0 & z_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & z_n \end{pmatrix} (G \setminus g_0^\tau)^\tau = g_0^\tau g_0 \quad (1)$$

is solvable over $\mathbb{F}_q$. Any solution $Z = (z_1, \dots, z_n)$ of (1) is a vector of recombination coefficients.

Proof: Define the $k \times n$ matrix

$$M := G \setminus g_0^\tau = (g_1^\tau, \dots, g_n^\tau),$$

and for a vector $Z = (z_1, \dots, z_n) \in \mathbb{F}_q^n$ consider the diagonal matrix

$$D_Z := \text{diag}(z_1, \dots, z_n) \in \mathbb{F}_q^{n \times n}.$$

Suppose first that the linear code C is multiplicative. By definition, this means that there exists $Z = (z_1, \dots, z_n)$ such that for every choice of secrets $s, s' \in \mathbb{F}_q$ and every corresponding choices of $u, u' \in \mathbb{F}_q^k$ with $s = ug_0^\tau$ and $s' = u'g_0^\tau$, the recombination identity

$$ss' = \sum_{i=1}^n z_i s_i s'_i$$

holds, where $s_i = ug_i^\tau$ and $s'_i = u'g_i^\tau$.

In terms of M and D_Z , this can be rewritten as

$$\sum_{i=1}^n z_i s_i s'_i = (s_1, \dots, s_n) D_Z (s'_1, \dots, s'_n)^\tau = u M D_Z M^\tau u'^\tau,$$

and

$$ss' = (ug_0^\tau)(u'g_0^\tau) = u(g_0^\tau g_0)u'^\tau.$$

Since the above equality

$$u M D_Z M^\tau u'^\tau = u(g_0^\tau g_0)u'^\tau$$

holds for *all* $u, u' \in \mathbb{F}_q^k$, we obtain an equality of bilinear forms on $\mathbb{F}_q^k \times \mathbb{F}_q^k$. It is well known that two bilinear forms over $\mathbb{F}_q^k$ are equal if and only if their Gram matrices coincide. Therefore,

$$M D_Z M^\tau = g_0^\tau g_0,$$

which is exactly the matrix equation in the statement of the theorem. Thus any recombination vector Z arising from a multiplicative ideal linear code necessarily gives a solution to that linear system.

Conversely, suppose that there exists $Z = (z_1, \dots, z_n) \in \mathbb{F}_q^n$ such that

$$M D_Z M^\tau = g_0^\tau g_0.$$

We claim that Z provides recombination coefficients for a multiplicative secret-sharing scheme based on C .

Indeed, let $s, s' \in \mathbb{F}_q$ be arbitrary secrets, and choose $u, u' \in \mathbb{F}_q^k$ such that $s = ug_0^\tau$ and $s' = u'g_0^\tau$. Let $uG = (s_0, s_1, \dots, s_n)$ and $u'G = (s'_0, s'_1, \dots, s'_n)$ be the corresponding codewords, and let s_i, s'_i be the resulting shares. Then

$$\sum_{i=1}^n z_i s_i s'_i = (s_1, \dots, s_n) D_Z (s'_1, \dots, s'_n)^\tau = u M D_Z M^\tau u'^\tau.$$

By the assumed matrix identity we have

$$u M D_Z M^\tau u'^\tau = u(g_0^\tau g_0)u'^\tau = (ug_0^\tau)(u'g_0^\tau) = ss'.$$

Since s, s' and the corresponding u, u' were arbitrary, this shows that for all pairs of secrets and for all randomness used in the distribution phase, the equality

$$ss' = \sum_{i=1}^n z_i s_i s'_i$$

holds. Hence C is a multiplicative ideal linear code with recombination coefficients $Z = (z_1, \dots, z_n)$. ■

We now derive a necessary condition for multiplicativity: the adversary structure induced by the target access structure must satisfy the Q_2 condition.

Theorem 2: For any access structure Γ , if an ideal linear code C of parameters $[n+1, k]$ satisfying $\Gamma(C) = \Gamma$ is multiplicative, then the corresponding adversary structure $\mathcal{A} = 2^P \setminus \Gamma$ satisfies the Q_2 condition [16]: that is, the union of any two unauthorized sets in $\mathcal{A}$ does not cover the entire participant set P .

Proof: Suppose that the adversary structure $\mathcal{A}$ does not satisfy the Q_2 condition. Then there exist $A_1, A_2 \in \mathcal{A}$ such that $A_1 \cup A_2 = P$. Let the generator matrix be

$$G = (g_0^\tau, g_1^\tau, \dots, g_n^\tau),$$

and let G_{A_1} denote the submatrix of G formed by the column vectors corresponding to the participants in A_1 . Set the secrets $s = s' = 1$, and let $u \in \mathbb{F}_q^k$ be used to distribute the shares via uG . Consider the linear system

$$ug_0^\tau = 1, \quad uG_{A_1} = 0. \quad (2)$$

Since $A_1 \in \mathcal{A}$, we have $A_1 \notin \Gamma$, which implies $g_0^\tau \notin \text{span}\{G_{A_1}\}$. Hence the system (2) has a solution u , and all shares corresponding to the secret $s = 1$ held by the participants in A_1 are zero.

Similarly, there exists $u' \in \mathbb{F}_q^k$ such that

$$u'g_0^\tau = s' = 1, \quad u'G_{A_2} = 0,$$

so that all shares corresponding to the secret s' held by the participants in A_2 are also zero. Therefore,

$$\sum_{i=1}^n z_i s_i s'_i = \sum_{P_i \in A_1} z_i s_i s'_i + \sum_{P_i \in A_2} z_i s_i s'_i = 0 \neq ss' = 1,$$

which contradicts the assumption that the linear code C is multiplicative. ■

Example 1: Consider the access structure

$$\begin{aligned} \Gamma = \{ & \{P_1, P_2, P_3\}, \{P_2, P_3, P_4\}, \{P_3, P_4, P_1\}, \{P_4, P_1, P_2\}, \\ & \{P_1, P_2, P_5\}, \{P_2, P_3, P_6\}, \{P_3, P_4, P_5\}, \{P_4, P_1, P_6\}, \\ & \{P_1, P_5, P_3\}, \{P_1, P_6, P_3\}, \{P_2, P_6, P_4\}, \{P_2, P_5, P_4\}, \\ & \{P_1, P_5, P_6\}, \{P_3, P_5, P_6\}, \{P_2, P_5, P_6\}, \{P_4, P_5, P_6\} \}. \end{aligned}$$

Let $\mathcal{A} = 2^P \setminus \Gamma$. Then

$$\mathcal{A} = \{ \{P_1, P_3\}, \{P_2, P_4\}, \{P_5, P_6\}, \{P_1, P_2, P_6\}, \{P_2, P_3, P_5\}, \{P_3, P_4, P_6\}, \{P_4, P_1, P_5\} \}.$$

It is easy to verify that $\mathcal{A}$ satisfies the Q_2 condition.

Let the generator matrix G be given by

$$G = \begin{pmatrix} 3 & 1 & 0 & 0 & 1 & 0 & 1 \\ 2 & 0 & 1 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 & 1 & 0 \end{pmatrix} = (g_0^\tau, g_1^\tau, g_2^\tau, g_3^\tau, g_4^\tau, g_5^\tau, g_6^\tau).$$

It is straightforward to verify that the linear code C over $\mathbb{F}_7$ generated by G satisfies $\Gamma(C) = \Gamma$. i.e., C is the ideal linear code that realizes Γ . Substituting G into the system

$$(G \setminus g_0^\tau) \begin{pmatrix} z_1 & 0 & \cdots & 0 \\ 0 & z_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & z_6 \end{pmatrix} (G \setminus g_0^\tau)^\tau = g_0^\tau g_0,$$

we obtain the following system of equations over $\mathbb{F}_7$:

$$\left\{ \begin{array}{l} z_1 + z_4 + z_6 = 2, \\ z_4 + z_6 = 6, \\ z_4 = 3, \\ z_2 + z_4 + z_5 + z_6 = 4, \\ z_4 + z_5 = 2, \\ z_3 + z_4 + z_5 = 1. \end{array} \right.$$

One feasible solution is

$$Z = (z_1, z_2, z_3, z_4, z_5, z_6) = (3, 6, 6, 3, 6, 3).$$

One can verify that

$$ss' = \sum_{i=1}^6 z_i s_i s'_i$$

holds for all $s, s' \in \mathbb{F}_7$, which implies that the ideal linear code C with generator matrix G is multiplicative.

IV. SHORTEST LINEAR CODES AND LENGTH-EFFICIENT CONSTRUCTIONS

The study of perfect secret sharing schemes began with Benaloh and Leichter [14], who discovered that for certain access structures, none of the secret sharing schemes that realize these access structures are ideal, and thus no ideal linear code C can realize these structures. Therefore, for access structures for which an ideal linear code is either non-existent or difficult to obtain, it is natural to consider a shortest linear code realizing the structure as a conceptual benchmark.

We next quantify the efficiency of a linear-code-based secret sharing scheme in terms of its total share size.

Definition 3 (Total share size): Let C be a linear code over $\mathbb{F}_q$ with parameters $[1 + \sum_{i=1}^n t_i, k]$ and generator matrix $G = (g_0^T, G_1, \dots, G_n)$, where $G_i \in \mathbb{F}_q^{k \times t_i}$ and $S_i = uG_i \in \mathbb{F}_q^{t_i}$ is the share vector of participant P_i . The *total share size* of C is defined as

$$\sigma(C) = \sum_{i=1}^n t_i,$$

i.e., the total number of field elements distributed as shares to all participants.

Definition 4 (Shortest linear code realizing Γ): Let Γ be an access structure. A linear code C over $\mathbb{F}_q$ with parameters $[1 + \sum_{i=1}^n t_i, k]$ and induced access structure $\Gamma(C) = \Gamma$ is called a *shortest linear code* realizing Γ if there is no linear code $\tilde{C}$ over $\mathbb{F}_q$ with parameters $[1 + \sum_{i=1}^n \tilde{t}_i, k]$ and $\Gamma(\tilde{C}) = \Gamma$ such that

$$\sigma(\tilde{C}) = \sum_{i=1}^n \tilde{t}_i < \sum_{i=1}^n t_i = \sigma(C).$$

In principle, one may regard shortest linear codes, in the sense of minimizing the total share size $\sigma(C)$ among all linear codes realizing a given access structure Γ , as an ideal benchmark for efficiency. However, explicitly determining such optimal codes for general access structures appears to be computationally infeasible. Desmedt, Kurosawa, and Van Le showed that even verifying whether a given linear scheme realizes a prescribed access structure is coNP-complete [17]. More recently, Applebaum and Nir proved that distinguishing between “cheap” access structures and “expensive” ones is coNP-hard [18], where “cheap” means admitting secret sharing schemes with small share size, and “expensive” means that any such scheme necessarily requires large shares. These results strongly suggest that for arbitrary access structures, exactly computing a linear code that minimizes the total share size is, in general, unlikely to be achievable in polynomial time.

Motivated by these complexity barriers, we retain shortest linear codes only as a theoretical benchmark and turn to a more practical goal: for a given access structure Γ , to construct linear codes whose induced access structure coincides with Γ while keeping the total share size $\sigma(C)$ comparatively small. To this end, we propose an explicit construction over $\mathbb{F}_q$, and refer to the resulting codes as *length-efficient linear codes* realizing Γ .

Construction 1 (length-efficient linear codes realizing Γ): Let the participant set be $P = \{P_1, \dots, P_n\}$, and let an access structure $\Gamma = \{S_1, \dots, S_m\}$ be given, where $S_i \subseteq P$ for $1 \leq i \leq m$. Define the following parity-check matrix $H \in \mathbb{F}_q^{m \times (n+1)}$:

$$H = \begin{pmatrix} 1 & h_{11} & h_{12} & \cdots & h_{1n} \\ 1 & h_{21} & h_{22} & \cdots & h_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & h_{m1} & h_{m2} & \cdots & h_{mn} \end{pmatrix} = \begin{pmatrix} h_1 \\ h_2 \\ \vdots \\ h_m \end{pmatrix},$$

where $h_{ij} = 1$ if $P_j \in S_i$, and $h_{ij} = 0$ otherwise, for $1 \leq i \leq m$ and $1 \leq j \leq n$.

1) Step 1 (Compute all codewords of the dual code C^* whose first coordinate equals 1):

Note that all codewords of the dual code C^* are linear combinations of the row vectors h_i , $1 \leq i \leq m$. Let $a_1, a_2, \dots, a_m \in \mathbb{F}_q$. For any $c^* \in C^*$, we have

$$c^* = a_1 h_1 + a_2 h_2 + \cdots + a_m h_m = (a_1 + \cdots + a_m, a_1 h_{11} + \cdots + a_m h_{m1}, \dots, a_1 h_{1n} + \cdots + a_m h_{mn}).$$

By imposing the constraint $a_1 + \cdots + a_m = 1$, we obtain all codewords of C^* over $\mathbb{F}_q$ whose first coordinate is equal to 1.

2) Step 2 (Determine the new minimal authorized sets):

Initialize an empty set AS_Γ . For every subset $A \subseteq P$, if there exists some $S_i \in \Gamma$ such that $S_i \subseteq A$, then discard A ; otherwise, add A to AS_Γ . Each element $A \in AS_\Gamma$ is called a *candidate new minimal authorized set*.

For each candidate new minimal authorized set $A \in AS_\Gamma$, set $\bar{A} = P \setminus A$. Without loss of generality, write

$$\bar{A} = \{P_{j_1}, P_{j_2}, \dots, P_{j_t}\}.$$

Consider the following system in the unknowns $a_1, a_2, \dots, a_m$:

$$\begin{cases} a_1 + \dots + a_m = 1, \\ a_1 h_{1j_1} + a_2 h_{2j_1} + \dots + a_m h_{mj_1} = 0, \\ a_1 h_{1j_2} + a_2 h_{2j_2} + \dots + a_m h_{mj_2} = 0, \\ \vdots \\ a_1 h_{1j_t} + a_2 h_{2j_t} + \dots + a_m h_{mj_t} = 0, \end{cases}$$

together with the inequalities

$$a_1 h_{1j} + a_2 h_{2j} + \dots + a_m h_{mj} \neq 0, \quad \text{for all } j \notin \{j_1, j_2, \dots, j_t\},$$

where all operations are performed over the finite field $\mathbb{F}_q$.

If this system admits a solution, then the candidate set A is indeed a new minimal authorized set generated by the parity-check matrix H , distinct from those in the original access structure, and A is retained in AS_Γ . Otherwise, if the system has no solution, the set A is removed from AS_Γ . The remaining elements of AS_Γ are precisely all new minimal authorized sets, and this procedure also yields their corresponding coefficients a_i ($1 \leq i \leq m$), which are clearly not all zero.

3) **Step 3 (Adding the minimum number of participant candidate columns):**

For each new minimal authorized set $A \in \text{AS}_\Gamma$, let a_i ($1 \leq i \leq m$) be the corresponding coefficients obtained in Step 2. Define

$$B_i = S_i \setminus (A \cap S_i), \quad 1 \leq i \leq m,$$

and

$$B = \{B_i \mid |B_i| = \min\{|B_1|, |B_2|, \dots, |B_m|\}\}.$$

If there are multiple B_i satisfying this condition, it suffices to choose one of them and denote it by B . The set B is then the minimal set of participants that needs to be added in order to eliminate the new minimal authorized set A . Without loss of generality, assume that a participant $P_j \in B$. Consider the $(j+1)$ -th column of the parity-check matrix H ,

$$(h_{1j}, h_{2j}, \dots, h_{mj})^\tau.$$

Whenever an entry h_{ij} is nonzero and the corresponding coefficient a_i for A is also nonzero, we add a candidate unit column vector e_i for the participant P_j , where $e_i \in \mathbb{F}_q^m$ has 1 in the i -th position and 0 in all other positions. This procedure is repeated until candidate columns have been added for all participants in B . If identical candidate columns would be added more than once for the same participant, they are added only once.

4) **Step 4 (Obtaining the length-efficient linear codes):**

After adding the candidate columns for the participants in Step 3, we obtain a new parity-check matrix H' . Let G' be a matrix satisfying

$$G'(H')^\tau = 0.$$

Then the linear code C' with generator matrix G' is the length-efficient linear codes that realizes the original access structure Γ .

We next illustrate Construction 1 on a concrete four-party access structure. This example shows how, starting from the minimal authorized sets of Γ , the construction produces a length-efficient linear code realizing Γ over $\mathbb{F}_{11}$.

Example 2: Consider the access structure

$$\Gamma = \{\{P_1, P_2\}, \{P_1, P_3\}, \{P_1, P_4\}, \{P_2, P_3, P_4\}\}$$

on the participant set $P = \{P_1, P_2, P_3, P_4\}$. Applying the above construction over $\mathbb{F}_{11}$ yields the shortest linear code that realizes Γ .

From the access structure Γ , we obtain the following parity-check matrix:

$$H = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} h_1 \\ h_2 \\ h_3 \\ h_4 \end{pmatrix},$$

Step 1. Any codeword $c^* \in C^*$ can be written as

$$c^* = a_1 h_1 + a_2 h_2 + a_3 h_3 + a_4 h_4,$$

TABLE I
CODEWORDS IN C^* AND COEFFICIENTS (a_1, a_2, a_3, a_4) CORRESPONDING TO NEW MINIMAL AUTHORIZED SETS

New minimal authorized set A	Codeword c^*	Coefficients (a_1, a_2, a_3, a_4)
$\{P_1\}$	$(1, 7, 0, 0, 0)$	$(6, 6, 6, 5)$
$\{P_2\}$	$(1, 0, 3, 0, 0)$	$(2, 10, 10, 1)$
$\{P_3\}$	$(1, 0, 0, 3, 0)$	$(10, 2, 10, 1)$
$\{P_4\}$	$(1, 0, 0, 0, 3)$	$(10, 10, 2, 1)$
$\{P_2, P_3\}$	$(1, 0, x_2, x_3, 0)$	$(a_1, a_2, 10, 1), (a_1 + a_2 = 1)$
$\{P_2, P_4\}$	$(1, 0, x_2, 0, x_4)$	$(a_1, 10, a_3, 1), (a_1 + a_3 = 1)$
$\{P_3, P_4\}$	$(1, 0, 0, x_3, x_4)$	$(10, a_2, a_3, 1), (a_2 + a_3 = 1)$

with $a_1, a_2, a_3, a_4 \in \mathbb{F}_{11}$, and hence

$$c^* = (a_1 + a_2 + a_3 + a_4, a_1 + a_2 + a_3, a_1 + a_4, a_2 + a_4, a_3 + a_4).$$

Imposing the constraint $a_1 + a_2 + a_3 + a_4 = 1$, we obtain all codewords of C^* over $\mathbb{F}_{11}$ whose first coordinate is equal to 1.

Step 2. The only candidate new minimal authorized sets (if any) are

$$\{P_1\}, \{P_2\}, \{P_3\}, \{P_4\}, \{P_2, P_3\}, \{P_2, P_4\}, \{P_3, P_4\}.$$

We first consider the candidate new minimal authorized set $\{P_1\}$. In order for the dual code C^* to generate the authorized set $\{P_1\}$, the corresponding codeword must be of the form

$$c^* = (1, x_1, 0, 0, 0), \quad x_1 \in \mathbb{F}_{11}, \quad x_1 \neq 0.$$

Consider the following system of equations:

$$\begin{cases} a_1 + a_2 + a_3 + a_4 = 1, \\ a_1 + a_4 = 0, \\ a_2 + a_4 = 0, \\ a_3 + a_4 = 0, \end{cases} \quad a_1 + a_2 + a_3 \neq 0,$$

where all operations are performed over F_{11} . Solving this system yields $a_1 = a_2 = a_3 = 6$ and $a_4 = 5$. Hence

$$x_1 = a_1 + a_2 + a_3 = 18 \equiv 7 \pmod{11},$$

and therefore there exists a new minimal codeword in the dual code,

$$c^* = 6h_1 + 6h_2 + 6h_3 + 5h_4 = (1, 7, 0, 0, 0),$$

which generates the new minimal authorized set $\{P_1\}$.

Next, we consider the candidate new minimal authorized set $\{P_2, P_3\}$. In order for the dual code C^* to generate the authorized set $\{P_2, P_3\}$, the corresponding codeword must be of the form

$$c^* = (1, 0, x_2, x_3, 0), \quad x_2, x_3 \in \mathbb{F}_{11}, \quad x_2 \neq 0, \quad x_3 \neq 0.$$

Consider the following system of equations:

$$\begin{cases} a_1 + a_2 + a_3 + a_4 = 1, \\ a_1 + a_2 + a_3 = 0, \\ a_3 + a_4 = 0, \end{cases} \quad a_1 + a_4 \neq 0, \quad a_2 + a_4 \neq 0,$$

again over $\mathbb{F}_{11}$. Solving this system yields $a_3 = 10$, $a_4 = 1$, and $a_1 + a_2 = 1$. It follows that

$$x_2 + x_3 = (a_1 + a_4) + (a_2 + a_4) = 3.$$

Hence all codewords in C^* that generate the authorized set $\{P_2, P_3\}$ are of the form

$$c^* = (1, 0, x_2, x_3, 0), \quad x_2, x_3 \in \mathbb{F}_{11}, \quad x_2 \neq 0, \quad x_3 \neq 0, \quad x_2 + x_3 = 3.$$

Similarly, the codewords corresponding to the remaining new minimal authorized sets, together with the associated coefficients of h_1, h_2, h_3, h_4 , are summarized in Table I.

Step 3. It is readily verified that

$$\text{AS}_\Gamma = \{\{P_1\}, \{P_2\}, \{P_3\}, \{P_4\}, \{P_2, P_3\}, \{P_2, P_4\}, \{P_3, P_4\}\},$$

and the feasible addition options corresponding to each new minimal authorized set are systematically listed in Table II.

TABLE II
FEASIBLE ADDITION OPTIONS FOR NEW MINIMAL AUTHORIZED SETS

New minimal authorized set A	Minimal set of participants to be added	Feasible candidate columns
$\{P_1\}$	$\{P_2\} \text{ or } \{P_3\} \text{ or } \{P_4\}$	$P_2 : \{e_1\} \text{ or } \{e_3\}; P_3 : \{e_2\} \text{ or } \{e_4\}; P_4 : \{e_3\} \text{ or } \{e_4\}$
$\{P_2\}$	$\{P_1\}$	$P_1 : \{e_1\} \text{ or } \{e_2\} \text{ or } \{e_3\}$
$\{P_3\}$	$\{P_1\}$	$P_1 : \{e_1\} \text{ or } \{e_2\} \text{ or } \{e_3\}$
$\{P_4\}$	$\{P_1\}$	$P_1 : \{e_1\} \text{ or } \{e_2\} \text{ or } \{e_3\}$
$\{P_2, P_3\}$	$\{P_1\} \text{ or } \{P_4\}$	$P_1 : \{e_3\}; P_4 : \{e_3\} \text{ or } \{e_4\}$
$\{P_2, P_4\}$	$\{P_1\} \text{ or } \{P_3\}$	$P_1 : \{e_2\}; P_3 : \{e_2\} \text{ or } \{e_4\}$
$\{P_3, P_4\}$	$\{P_1\} \text{ or } \{P_2\}$	$P_1 : \{e_1\}; P_2 : \{e_1\} \text{ or } \{e_4\}$

Selection rule: If identical candidate columns would be added more than once for the same participant, they are added only once.

Based on the above table, we finally select the candidate columns e_1 and e_3 for participant P_1 , and the candidate column e_1 for participant P_2 . Although this choice does not explicitly include the feasible candidate columns associated with the new minimal authorized set $\{P_2, P_4\}$, such as e_2 for P_1 , the above combination already suffices to eliminate $\{P_2, P_4\}$. Indeed, for any valid set of coefficients associated with the authorized set $\{P_2, P_4\}$, the coefficients a_1 and a_3 satisfy $a_1 + a_3 = 1$, and hence cannot both be zero. Consequently, at least one of the added columns e_1 and e_3 contributes a nonzero component to the corresponding linear combination for this authorized set, so that it is no longer a new minimal authorized set. Therefore, there is no need to add e_2 .

Step 4. After adding the candidate columns as described in Step 3, we obtain the new parity-check matrix

$$H' = \begin{pmatrix} 1 & 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix},$$

and a corresponding generator matrix

$$G' = \begin{pmatrix} 1 & 10 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 10 & 10 & 0 & 0 & 1 & 0 \\ 1 & 10 & 0 & 1 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

The column indices are mapped to participants according to

$$\varphi = \begin{pmatrix} 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ P_1 & P_1 & P_1 & P_2 & P_2 & P_3 & P_4 \end{pmatrix}.$$

Thus, the linear code C' with generator matrix G' is the length-efficient linear code that realizes the original access structure Γ .

V. MULTIPLICATIVE LENGTH-EFFICIENT LINEAR CODES

In this section, we first consider a general linear-code-based secret-sharing scheme. Let C be a linear code over $\mathbb{F}_q$ with generator matrix

$$G = (g_0^\tau, G_1, \dots, G_n),$$

where $G_i \in \mathbb{F}_q^{k \times t_i}$ for $1 \leq i \leq n$, and assume that the secret-sharing scheme induced by C realizes an access structure Γ on the participant set $P = \{P_1, \dots, P_n\}$. The following description of secret distribution, secret reconstruction, and additive homomorphism applies to any such linear code C ; in particular, it applies to the length-efficient linear codes obtained from Construction 1.

1) Secret distribution: Let $s \in \mathbb{F}_q$ be the secret. Choose $u \in \mathbb{F}_q^k$ uniformly at random such that $ug_0^\tau = s$, and set $uG = (s_0, S_1, \dots, S_n)$, where $s_0 = s$ and $S_i = uG_i \in \mathbb{F}_q^{t_i}$ is a t_i -dimensional row vector, written as $S_i = (S_{i1}, S_{i2}, \dots, S_{it_i})$, whose components serve as the t_i shares of participant P_i .

2) Secret reconstruction: Let $A = \{P_1, \dots, P_m\} \subseteq \Gamma$ be an authorized set. The participants in A can reconstruct the secret s if and only if there exist vectors $u_i \in \mathbb{F}_q^{t_i}$ for $P_i \in A$ such that

$$g_0^\tau = \sum_{i=1}^m G_i u_i^\tau.$$

In this case,

$$s = ug_0^\tau = \sum_{i=1}^m uG_i u_i^\tau = \sum_{i=1}^m S_i u_i^\tau.$$

Therefore, the access structure realized by this scheme is

$$\Gamma = \{ A \subseteq P \mid g_0 \in \text{span}\{G_i \mid P_i \in A\} \}.$$

3) Additive homomorphism: Similarly, for secrets $s, s' \in \mathbb{F}_q$ and their corresponding share vectors S_i and S'_i (for $1 \leq i \leq n$), one obtains

$$s + s' = ug_0^\tau + u'g_0^\tau = \sum_{i=1}^n S_i u_i^\tau + \sum_{i=1}^n S'_i u_i^\tau = \sum_{i=1}^n (S_i + S'_i) u_i^\tau.$$

Thus, the participants only need to compute the sums of their local share vectors $S_i + S'_i$, after which an authorized set can jointly reconstruct $s + s'$.

However, the computation of the product ss' is more involved:

$$ss' = \left(\sum_{i=1}^n S_i u_i^\tau \right) \left(\sum_{i=1}^n S'_i u_i^\tau \right).$$

A direct expansion produces numerous undesired cross terms. It is therefore desirable that multiplication can be realized solely from local products of each participant's own shares; that is, there exist matrices $Z_i \in \mathbb{F}_q^{t_i \times t_i}$ for $1 \leq i \leq n$ such that

$$ss' = \sum_{i=1}^n S_i Z_i (S'_i)^\tau.$$

Motivated by this objective, We now specialize to the length-efficient linear codes obtained from Construction 1 and introduce multiplicativity in this setting.

Definition 5 (Multiplicative length-efficient linear code): Let Γ be an access structure, and let C be a length-efficient linear code over $\mathbb{F}_q$ realizing Γ , obtained from Construction 1. Suppose C has parameters $[1 + \sum_{i=1}^n t_i, k]$ and generator matrix

$$G = (g_0^\tau, G_1, \dots, G_n),$$

where $G_i \in \mathbb{F}_q^{k \times t_i}$ for $1 \leq i \leq n$, and the share vector of participant P_i is

$$S_i = uG_i = (S_{i1}, \dots, S_{it_i}) \in \mathbb{F}_q^{t_i}.$$

If there exist matrices $Z_i \in \mathbb{F}_q^{t_i \times t_i}$ for $1 \leq i \leq n$ such that for all secrets $s, s' \in \mathbb{F}_q$ (with corresponding share vectors S_i, S'_i generated as above) one has

$$ss' = \sum_{i=1}^n S_i Z_i (S'_i)^\tau,$$

then C is called a *multiplicative length-efficient linear code* realizing Γ .

Theorem 3: Let C be a length-efficient linear code with parameters $[1 + \sum_{i=1}^n t_i, k]$ and generator matrix

$$G = (g_0^\tau, G_1, G_2, \dots, G_n),$$

where $G_i \in \mathbb{F}_q^{k \times t_i}$ for $1 \leq i \leq n$. Then C is a multiplicative length-efficient linear code if and only if the linear system

$$(G \setminus g_0^\tau) Z (G \setminus g_0^\tau)^\tau = g_0^\tau g_0 \quad (3)$$

has a solution, where Z is the block diagonal matrix

$$Z = \begin{pmatrix} Z_1 & 0 & \cdots & 0 \\ 0 & Z_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & Z_n \end{pmatrix},$$

and each diagonal block satisfies $Z_i \in \mathbb{F}_q^{t_i \times t_i}$ for $1 \leq i \leq n$.

Proof: Assume that the system (3) has a solution. Let $s, s' \in \mathbb{F}_q$ be two secrets, and let $u, u' \in \mathbb{F}_q^k$ be the corresponding vectors used in the share distribution, so that $s = ug_0^\tau$ and $s' = u'g_0^\tau$, and $S_i = uG_i$, $S'_i = u'G_i$ for $1 \leq i \leq n$. Then

$$\begin{aligned} \sum_{i=1}^n S_i Z_i (S'_i)^\tau &= \sum_{i=1}^n (uG_i) Z_i (u'G_i)^\tau \\ &= u(G_1, \dots, G_n) Z (G_1, \dots, G_n)^\tau (u')^\tau \\ &= u(G \setminus g_0^\tau) Z (G \setminus g_0^\tau)^\tau (u')^\tau \\ &= u(g_0^\tau g_0) (u')^\tau \\ &= (ug_0^\tau) (g_0 (u')^\tau) \\ &= ss'. \end{aligned}$$

Hence C satisfies

$$ss' = \sum_{i=1}^n S_i Z_i (S'_i)^\tau$$

for all $s, s' \in F_q$, and therefore C is a multiplicative length-efficient linear code.

Conversely, suppose that C is a multiplicative length-efficient linear code. Then there exist matrices $Z_i \in \mathbb{F}_q^{t_i \times t_i}$, $1 \leq i \leq n$, such that

$$ss' = \sum_{i=1}^n S_i Z_i (S'_i)^\tau$$

for all $s, s' \in \mathbb{F}_q$. Define the block diagonal matrix

$$Z = \begin{pmatrix} Z_1 & 0 & \cdots & 0 \\ 0 & Z_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & Z_n \end{pmatrix}.$$

Using the same computation as above, we obtain

$$u(G \setminus g_0^\tau) Z (G \setminus g_0^\tau)^\tau (u')^\tau = u(g_0^\tau g_0) (u')^\tau$$

for all $u, u' \in \mathbb{F}_q^k$, which implies

$$(G \setminus g_0^\tau) Z (G \setminus g_0^\tau)^\tau = g_0^\tau g_0.$$

Thus Z is a solution to (3). ■

Similarly, by an argument analogous to the proof of Theorem 2, it follows that if a length-efficient linear code C is multiplicative, then the corresponding adversary structure $\mathcal{A}$ satisfies the Q_2 condition.

Example 3: Consider the access structure

$$\Gamma = \{\{P_1, P_2\}, \{P_1, P_3\}, \{P_1, P_4\}, \{P_2, P_3, P_4\}\}$$

on the participant set $P = \{P_1, P_2, P_3, P_4\}$. It can be verified that the adversary structure corresponding to Γ satisfies the Q_2 condition.

We now verify that the length-efficient linear code C' over $\mathbb{F}_{11}$ obtained in Example 2, which realizes the access structure Γ , is multiplicative. The generator matrix of C' is given by

$$G' = \begin{pmatrix} 1 & 10 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 10 & 10 & 0 & 0 & 1 & 0 \\ 1 & 10 & 0 & 1 & 0 & 0 & 0 & 1 \end{pmatrix} = (g_0^\tau, G_1, G_2, G_3, G_4),$$

where G_1 is a 4×3 matrix, G_2 is a 4×2 matrix, and G_3 and G_4 are 4×1 matrices. In this case, C' is a linear code with parameters $[8, 4]$.

Let

$$Z_1 = \begin{pmatrix} z_1 & z_2 & z_3 \\ z_4 & z_5 & z_6 \\ z_7 & z_8 & z_9 \end{pmatrix}, \quad Z_2 = \begin{pmatrix} z_{10} & z_{11} \\ z_{12} & z_{13} \end{pmatrix}, \quad Z_3 = (z_{14}), \quad Z_4 = (z_{15}),$$

and consider the block diagonal matrix

$$Z = \begin{pmatrix} Z_1 & 0 & 0 & 0 \\ 0 & Z_2 & 0 & 0 \\ 0 & 0 & Z_3 & 0 \\ 0 & 0 & 0 & Z_4 \end{pmatrix}.$$

Substituting Z into the linear system

$$(G' \setminus g_0^\tau) Z (G' \setminus g_0^\tau)^\tau = g_0^\tau g_0,$$

we obtain the following system of linear equations over $\mathbb{F}_{11}$:

$$\left\{ \begin{array}{l} z_1 + 10z_4 + 10z_2 + z_5 + z_{10} = 1, \\ 10z_2 + z_5 + z_{11} = 0, \\ z_2 + 10z_5 + z_3 + 10z_6 = 1, \\ z_1 + 10z_4 + 10z_3 + z_6 = 1, \\ 10z_4 + z_5 + z_{12} = 0, \\ z_5 + z_{13} = 0, \\ 10z_5 + 10z_6 = 0, \\ 10z_4 + z_6 = 0, \\ z_4 + z_7 + 10z_5 + 10z_8 = 1, \\ 10z_5 + 10z_8 = 0, \\ z_5 + z_8 + z_6 + z_9 + z_{14} = 1, \\ z_4 + z_7 + 10z_6 + 10z_9 = 1, \\ z_1 + 10z_7 + 10z_2 + z_8 = 1, \\ 10z_2 + z_8 = 0, \\ z_2 + 10z_8 + z_3 + 10z_9 = 1, \\ z_1 + 10z_7 + 10z_3 + z_9 + z_{15} = 1. \end{array} \right.$$

This system admits the solution

$$Z_1 = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad Z_2 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Z_3 = (1), \quad Z_4 = (1).$$

It can be verified that, over $\mathbb{F}_{11}$,

$$ss' = \sum_{i=1}^4 S_i Z_i (S'_i)^\tau,$$

and therefore the linear code C' with parameters $[8, 4]$ and generator matrix G' is multiplicative.

For the four-party access structure Γ considered above, every participant appears in some minimal authorized set. Hence any MSP computing the corresponding monotone Boolean function f_Γ must have at least four rows. Let t denote the row size of such an MSP. By the generic construction of Cramer *et al.* [6], starting from a t -row MSP, one obtains a multiplicative MSP for the same access structure Γ with $2t$ rows. In particular, for this access structure any multiplicative MSP obtained in this way has at least $2t \geq 8$ rows.

Furthermore, as shown in [19], for a fixed access structure Γ there is a one-to-one correspondence between MSPs over $\mathbb{F}_q$ that compute f_Γ and linear-code-based secret sharing schemes over $\mathbb{F}_q$ that realize Γ . In particular, an MSP with t rows corresponds to a linear code whose total share size equals t , and conversely. Therefore, for the access structure Γ under consideration, any multiplicative linear code obtained via the generic MSP-based construction of Cramer *et al.* must have total share size at least 8.

As a concrete instance, consider the MSP over $\mathbb{F}_{11}$ defined by

$$M = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \\ 2 & 1 & 0 \end{pmatrix},$$

and the labeling map φ given by

$$\varphi = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ P_1 & P_1 & P_2 & P_3 & P_4 \end{pmatrix}.$$

It can be verified that the MSP $\mathcal{M}(\mathbb{F}_{11}, M, \varphi, v)$ with target vector $v = (1, 0, 0)$ computes the monotone Boolean function f_Γ , but it is not multiplicative. Using the multiplicative MSP construction of Cramer *et al.*, one obtains a linear code C with parameters $[11, 5]$, whose total share size is 10.

In contrast, our multiplicative length-efficient linear code C' realizing the same access structure Γ has parameters $[8, 4]$ and total share size 7. Thus, for this access structure, the proposed length-efficient construction yields a multiplicative linear secret sharing scheme with strictly smaller total share size than that obtained by applying the generic MSP-based approach of Cramer *et al.*.

VI. CONCLUSION

In this paper, we have carried out a concrete study of linear-code-based secret sharing schemes with a focus on multiplicative properties. We introduced the notion of shortest linear codes and proposed an efficient construction that, for any given access structure, produces a length-efficient linear code realizing it. On this basis, we further defined multiplicative ideal linear codes and multiplicative length-efficient linear codes, derived explicit linear-algebraic criteria for multiplicativity, and demonstrated the effectiveness of the construction through examples over $\mathbb{F}_{11}$. For the access structures considered, the multiplicative linear codes obtained by our method have strictly smaller length than those arising from the generic MSP-based construction of Cramer *et al.* [6], leading to more efficient secret sharing schemes for secure multiparty computation.

Several questions remain open. A central one is whether, for any access structure that admits a multiplicative realization, the length-efficient linear code produced by our construction is always multiplicative, or at least can be made multiplicative by iteratively adding further candidate columns. It would also be interesting to sharpen upper and lower bounds on the achievable length of multiplicative linear codes for concrete families of access structures, and to improve the complexity and practical efficiency of the construction, for example by optimizing the choice of field size or by exploiting additional algebraic structure in the underlying codes.

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